Code Boilerplate generator (cbp-gen)

Project description

A native application that allows me to create boilerplate code for any programming language, C/C++, Go, React, and more in the future.

Requirements

- Must be able to run on all computer operating systems (MacOS, Unix/Linux, Windows)
- Should have a front-end that is capable of connecting with a back-end
- Should use a modern, fast language to quickly generate the boilerplate
- Must allow user inputs to customize the boilerplate code
- For minimum viable product, must be able to create boilerplate for these languages:
    - Go, C, C++, HTML, React, JavaScript
- Can run application from anywhere on the system and be able to create boilerplate code in specific correct locations
- Can access the terminal/command line to be able to run commands such as “npm install”.
- Allow for user customizability with what they want in their boilerplate code
- This application should be able to scale with multiple new languages for boiler plates

Agile Sprints

First Sprint:

- ~~Pick a programming language~~
- ~~Make a new file in the current directory~~
- ~~Make a file using the commandline/shell~~
- ~~Run a commandline/shell command~~
- ~~Detect if a file already exists~~
- ~~Detect which OS is benign run on the system~~
- ~~Learn how file paths work using Go and changing file paths~~

Second Sprint:

- ~~Implement a template format for C~~
- ~~Write a basic CLI styled program for the boilerplate generator~~
- ~~Learn how to make flags while using Go~~
- ~~Allow customization for user:~~

- ~~What type~~
- ~~Change the name~~

Third Sprint:

Due to implementing many features in Sprint 2, Sprint 3 will be update to reprioritize feature implementation

- Allow function arguments to be generated
- Add more programming languages (JavaScript, TypeScript, Python, Kotlin, Java, C#, Rust, PHP)
- Throw errors if unnecessary flags are added to the code
- If applicable, have a way to generate header files
- (OPTIONAL): If applicable, generate Class definitions instead of files

Fourth Sprint:

We will no longer use Weils since it offers features that are not needed given the scope of this project

- Decide on a GUI framework for the project using Go
- Make a GUI for the project:
  - Drop down menu to select language
  - Input fields for name, class, etc.
  - Buttons (radials) for any boolean operation
  - Input for the file path

Fifth Sprint:

- Update the GUI to be nicer.
- Make a website to host the project for people to be able to download